Fuse unless tense
Rhotic-vowel interaction in Elbe-Weser German

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Main Claim

- In Elbe-Weser German coda /ʁ/ vocalizes after tense vowels and fuses with lax vowels.
- We claim that this pattern can be derived if German /ʁ/ is specifed as [–tense].
- An OCP(tense) constraint enforces the different phonological processes.

Outline

Introduction

Data: Dissimilation and R-fusion

Analysis: OCP(tense)

Pilot Study: Lowering and Diphthongization

Conclusion

Introduction

- German coda /ʁ/ undergoes different processes after lax vowels and tense vowels (i.a. Hall 1992; Wiese 2000).
- Wiese (2000) already doubts consonantal realization and suspects neutralization of tenseness in this context for Northern German.
- We provide phonetic data from the Elbe-Weser variety of German.
- These data show that /ʁ/ **fuses** with lax vowels, but **vocalizes** after tense vowels.
- We will analyze this as an effect of an OCP constraint on tenseness, that is repaired by fusion or dissimlation.
- This supports the claim that German /ʁ/ is underlyingly specified as [–tense] (cf. Wiese 2000; Kostakis 2015).

Data

High tense vowels

- Coda-ʁ that follows a high, tense vowel is invariably turned into [ɐ̯], creating a diphthong.

(1)

/biːʁ/	→	[biːɐ̯]	'beer'
/tsiːʁ/	→	[tsiːɐ̯]	'decoration'
/tyːʁ/	→	[tyːɐ̯]	'door'
/kyːʁ/	→	[kyːɐ̯]	'freestyle figure skating'
/uːʁ/	→	[ʔuːɐ̯]	'clock'
/ʃtuːʁ/	→	[ʃtuːɐ̯]	'stubborn'

High lax round vowels

- Coda-ʁ following high lax vowels is not realised as [ɐ̯].
- Instead, the vowel is lowered and lengthened.

(2)

/dʏʁ/	→	[dœː]	‘meagre’
/mʏʁbə/	→	[mœːbə]	‘meagre’
/lʊʁç/	→	[lɔːç]	‘amphibian’
/kʊʁts/	→	[kɔːts]	‘short’

High lax non-round vowels

- Lax, unrounded vowel /ɪ/ is also considerably backed and optionally rounded.

(3)

/ɪʁ/	→	[ʔœː]∼[ʔə̈ː]∼[ʔə̈ː]	‘crazy’
/vɪʁ/	→	[vœː]∼[və̈ː]∼[və̈ː]	‘confused’
/hɪʁtə/	→	[hœː]∼[hə̈ːtə]∼[hə̈ːtə]	‘shepherd’

Mid tense vowels

- With mid tense vowels, coda-ʁ shows up mostly as [ɐ̯].
- If it does not, the vowel is laxed.

(4)

/feːʁd/	→	[feːɐ̯t] ∼ [fɛːt]	‘horse’
/heːʁdə/	→	[heːɐ̯də] ∼ [hɛːdə]	‘herd’
/ʃtøːʁ/	→	[ʃtøːɐ̯]∼ [ʃtœː]	‘sturgeon’
/ɛmpøːʁt/	→	[ʔɛmpøːɐ̯t]∼ [ʔɛmpœːt]	‘outraged’
/moːʁ/	→	[moːɐ̯]∼ [mɔː]	‘bog’
/koːʁ/	→	[koːɐ̯]∼ [kɔː]	‘choir’

Core data: Mid lax vowels

- If coda-ʁ does not show up as [ɐ̯], lax mid vowels are lengthened.
- If it does show up as [ɐ̯], lax mid vowels are tensed.

(5)

/hɛʁ/	→	[hɛː]∼[heːɐ̯]	'Sir'
/dœʁ/	→	[dœː]∼[døːɐ̯]	'dried-'
/nɔʁdn̩/	→	[nɔːdn̩]∼[noːɐ̯dn̩]	'north'

Low vowels

- After low vowel, coda-ʁ is never realised as [ɐ].
- Length distinction is completely neutralised.

(6)

/ba(ː)ʁ/	→	[baː]	‘bar’
/haːʁ/	→	[haː]	‘hair’
/naʁ/	→	[naː]	‘fool’

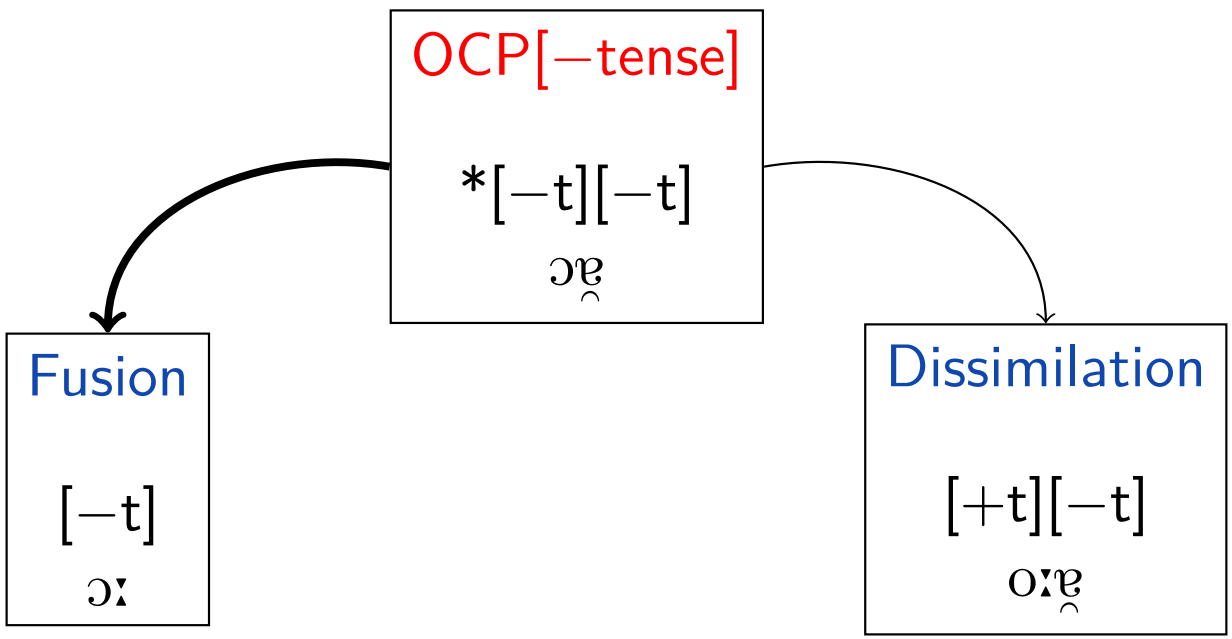
Analysis

Assumptions

- Optimality theory (Prince & Smolensky, 1993): Phonology evaluates possible outputs against a set of ranked constraints.
- Number of possible subrankings between partially ranked constraints predicts frequency of outputs in optional variation (Nagy & Reynolds, 1997; Anttila & Cho, 1998).
- Constraints can be restricted to a certain domain (Odden, 1994; Suzuki, 1998).
- Dorsal consonants can be specified with vocalic featues (Chomsky & Halle, 1968; Kostakis, 2015).

Gist

- ʁ and its vocalic counterpart ɐ̯ are [-t(ense)].
- OCP on [-tense] must be obeyed in the rime (cf. Suzuki 1998).
- Two ways to avoid it: fusion or dissimilation.



Constraints

Undominated markedness constraints:

- *ʁ]$_\sigma$ Assign $*$ to coda-ʁ.
- OCP(-t) Assign $*$ to adjacent [−tense] segments in a rime.

Unranked constraints:

- *œː Assign $*$ to [−low−tense] long vowels.
- Id(t) Assign $*$ to I-O correspondent segments with different values for [±tense].
- *Diph Assign $*$ to diphthongs.

Low ranked faithfulness constraint:

- Uni Assign $*$ for each many-to-one I-O mapping.

Lax mid vowels prefer fusion

- OCP(-t) blocks the faithful diphthong.
- 4 subrankings of the unranked constraints lead to the more likely optimal candidate: fusion.
- 2 subrankings lead to the less likely optimal candidate: dissimilation.

(7) /nɔ$_1$ʁ$_2$d/	*ʁ]$_\sigma$	OCP(-t)	*ʏː	Id(t)	*œː	*Diph	Uni
a. [nɔ$_1$ʁ$_2$t]	*!	*					
☞b. [nɔː$_{1,2}$t]					*		*
☞c. [nɔː$_1$e̯$_2$t]				*		*	
d. [nɔ$_1$e̯$_2$t]		*!		*		*	

Tense mid vowels prefer (vacuous) dissimilation

- 4 subrankings lead to the more likely optimal candidate: dissimilation.
- 2 subrankings lead to the less likely optimal candidate: fusion.

(8) /ʃtøː$_1$ʁ$_2$/	*ʁ]$_\sigma$	OCP(-t)	*yː	Id(t)	*œː	*Diph	Uni
a. [ʃtøː$_1$ʁ$_2$]	*!						
☞b. [ʃtœː$_{1,2}$]				*	*		*
☞c. [ʃtøː$_1$ɐ̯$_2$]						*	

Sub-rankings

(9) Possible rankings and outputs for mid vowels

/øːʁ/	*rankings*				/ɔʁ/
øːɐ̯	*œː	»	Id(t),	*Diph	oːɐ̯
	Id(t)	»	*œː,	*Diph	ɔː
œː	*Diph	»	*œː,	Id(t)	

High vowels

Additional undominated constraint:

- *ʏː Assign $*$ to [+high −tense] long vowels.

Tense high vowels must not fuse

- *ʏː blocks the fused candidate.

(10) /yːɪ$_1$ʁ$_2$/	*ʁ]$_\sigma$	OCP(-t)	*ʏː	Id(t)	*œː	*Diph	Uni
a. [yːɪ$_1$ʁ$_2$]	*!						
b. [ʏːɪ$_{1,2}$]			*!				*
☞c. [yːɪ$_1$e̯$_2$]						*	

Interim summary

- /ʁ/ cannot surface as vocalized [ɐ] after lax vowels due to an OCP(–tense) constraint.
- Mid vowels vary between a fused and a dissimilated form because crucial constraints are left unranked with respect to each other.
- Preferences are derived by the number of subrankings leading to this ouput.
- High vowels only fuse if lax in the input.

Pilot Study

Recordings

- 2 male participants aged 20-30 from the Elbe-Weser region (TW and HW).
- Recording inside a sound attenuated booth.
- Stimuli presented orthographically on a computer screen.
- 120 items in randomized order with a word containing /ʁ/ embedded inside a carrier sentence.
- Two conditions: Vowels before coda-R (Vr.) and vowels before onset-R (V.r).
- Balanced s.t. each vowel occured in both conditions.
- Each word repeated three times.

Annotation & Measurements

- For each of the high and mid vowels (138 tokens) the vocalic portion was annotated with Praat (Boersma & Weenink, 2017).
- Vocalic portion includes /ʁ/ in Vr. but excludes it in V.r
- First and second formant measured at first, second, and third quarter of vocalic portion.
- 6 tokens excluded because the participants either did not produce the stimulus at all or diverged in their pronunciation.

Effects of Coda-R

- High lax and mid tense vowels have a higher first formant and a slightly lower second formant in Vr. Condition (compared to V.r).
- No such effect for mid lax vowels.
- Neutralization of contrasts in Vr. condition.
- In the following figures: Vr. condition in brown and V.r condition in blue

Figure 1: TW high lax vowels

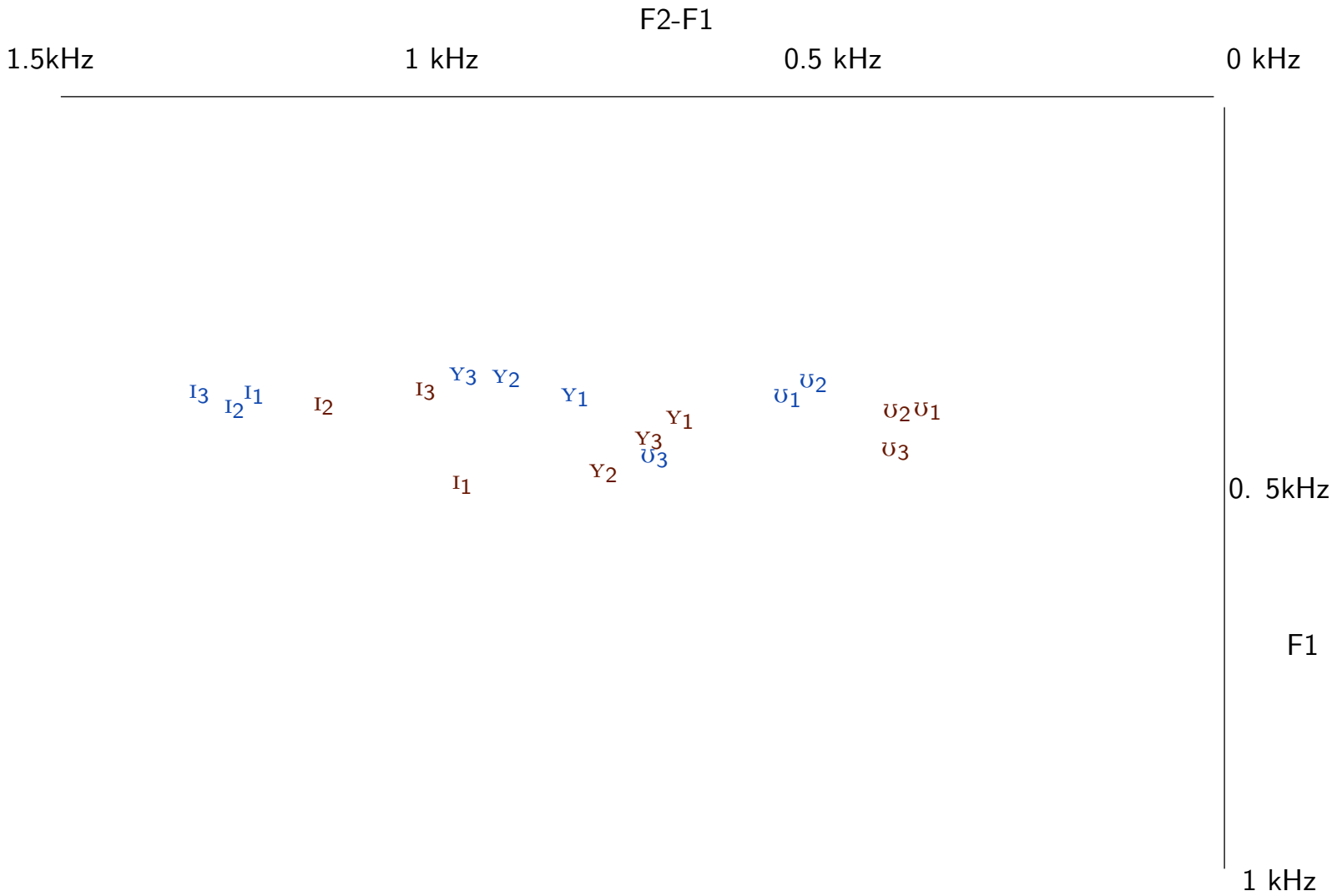


Figure 2: HW high lax vowels

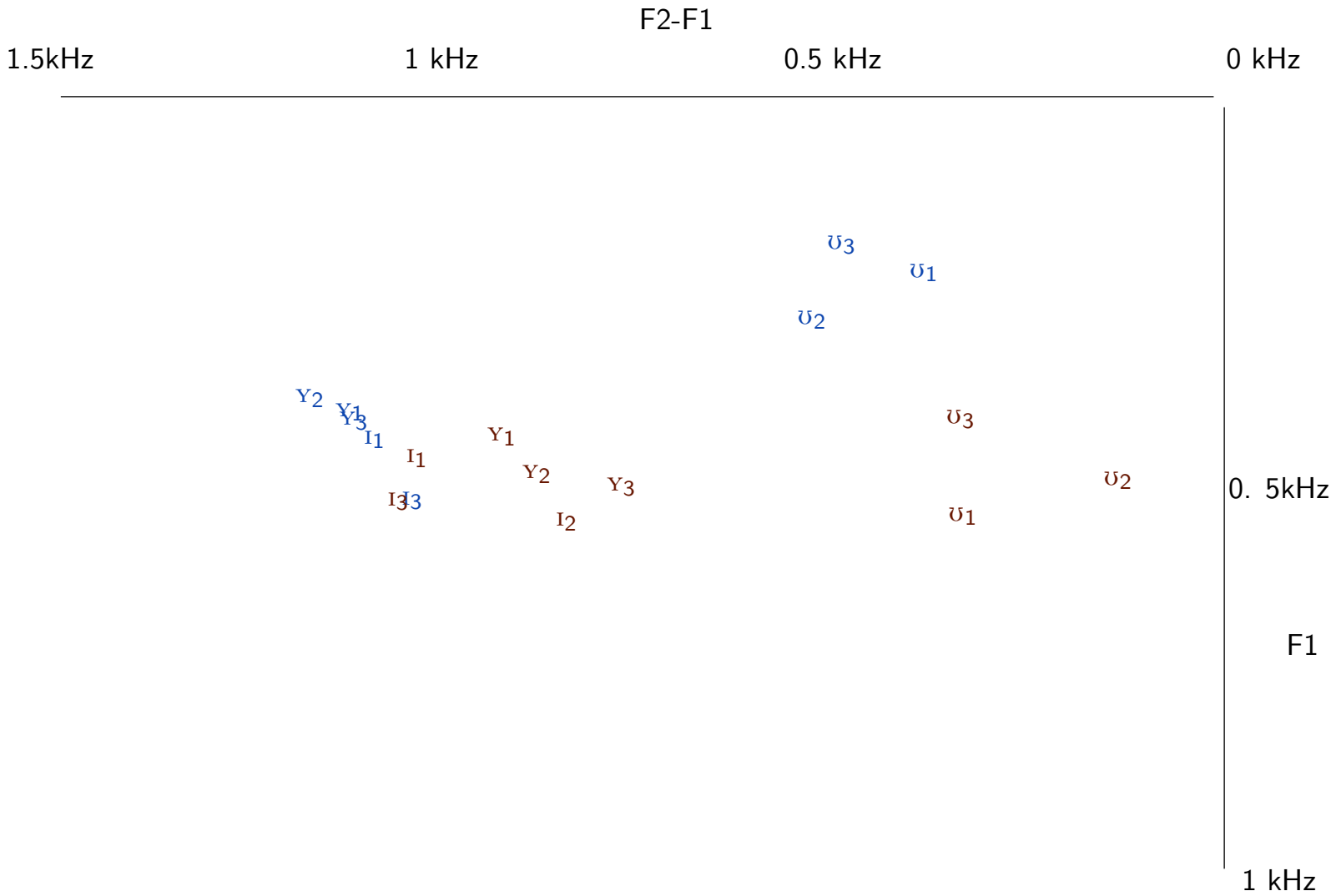


Figure 3: TW mid tense vowels

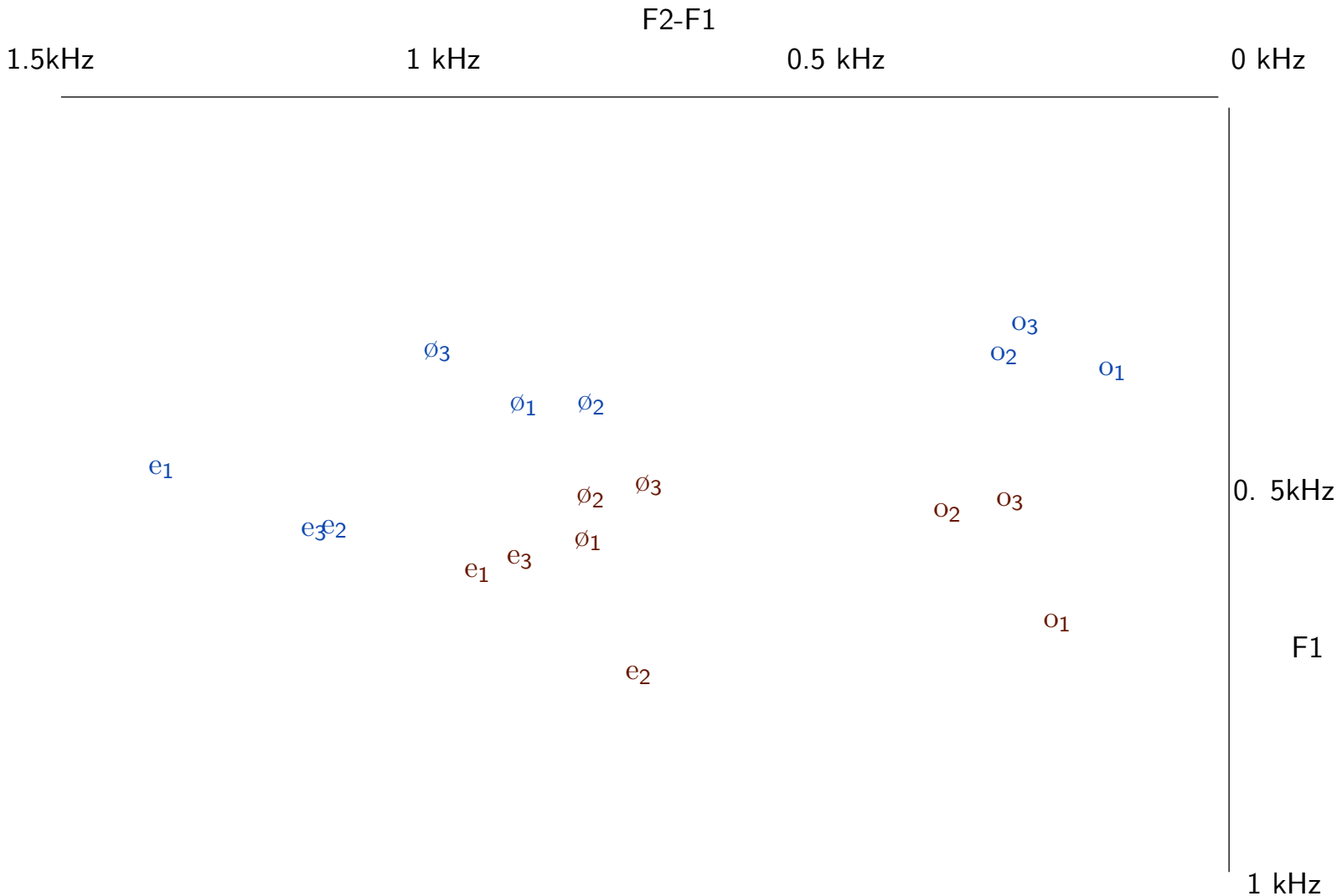


Figure 4: HW mid tense vowels

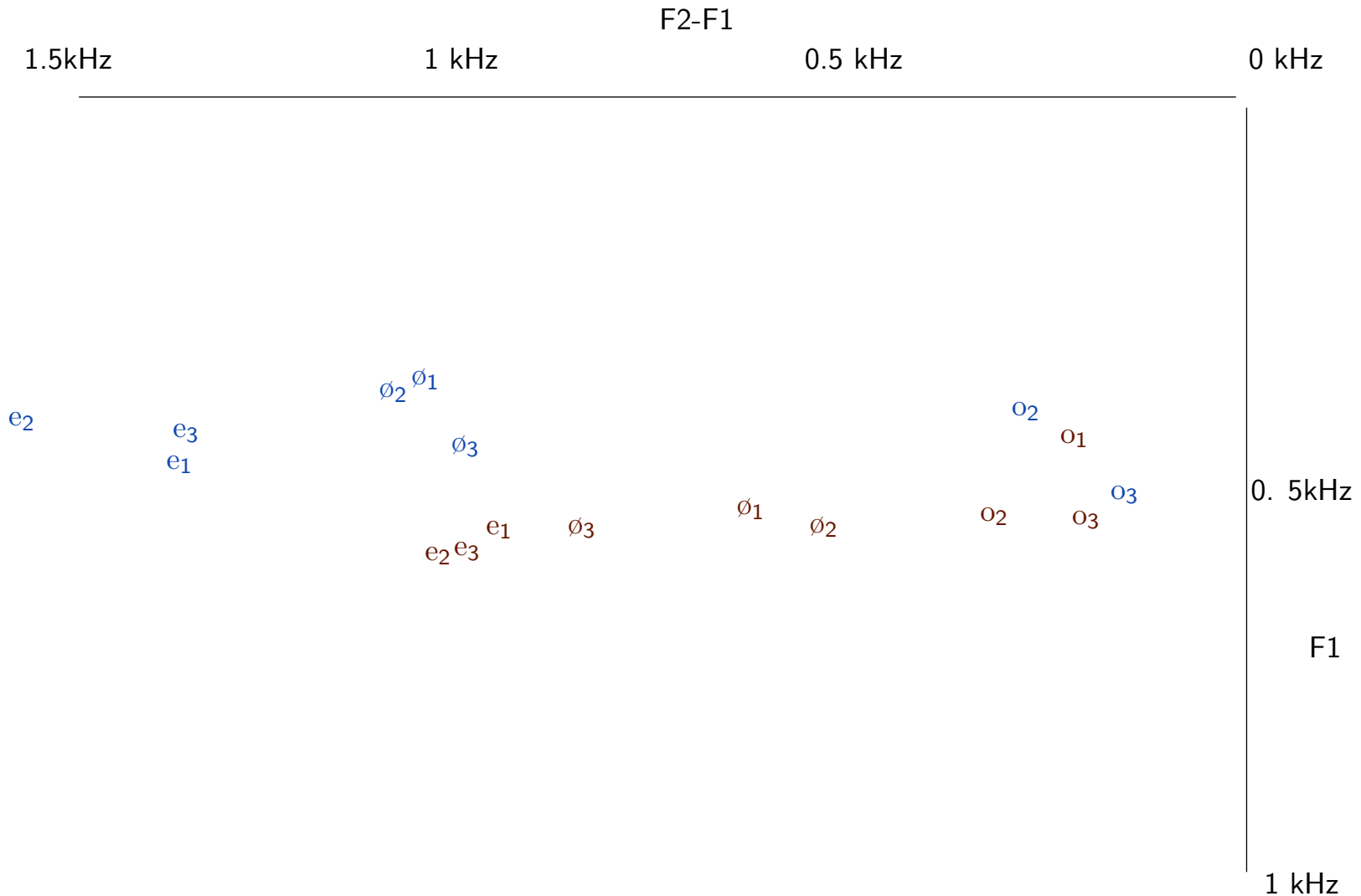


Figure 5: TW mid lax vowels

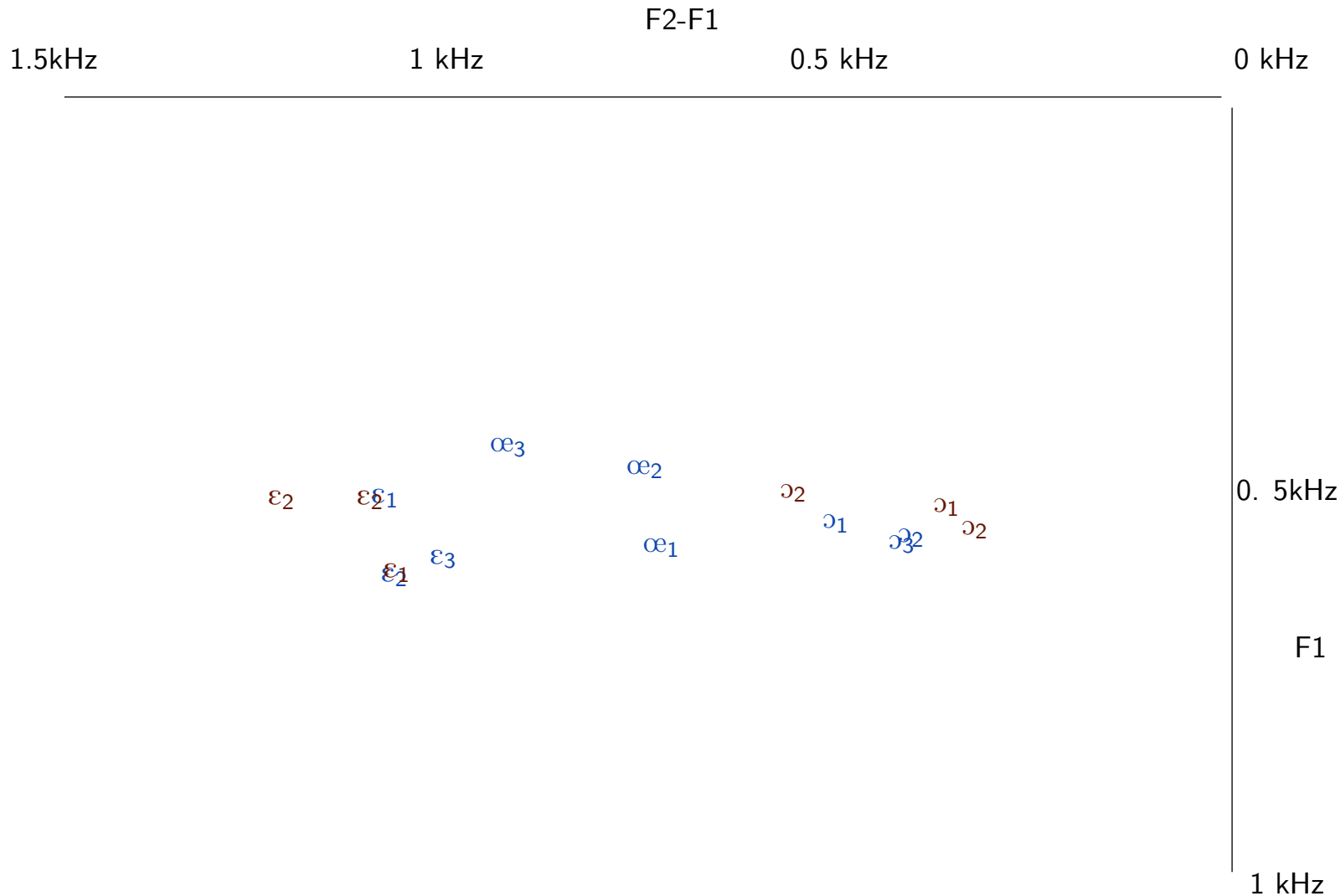
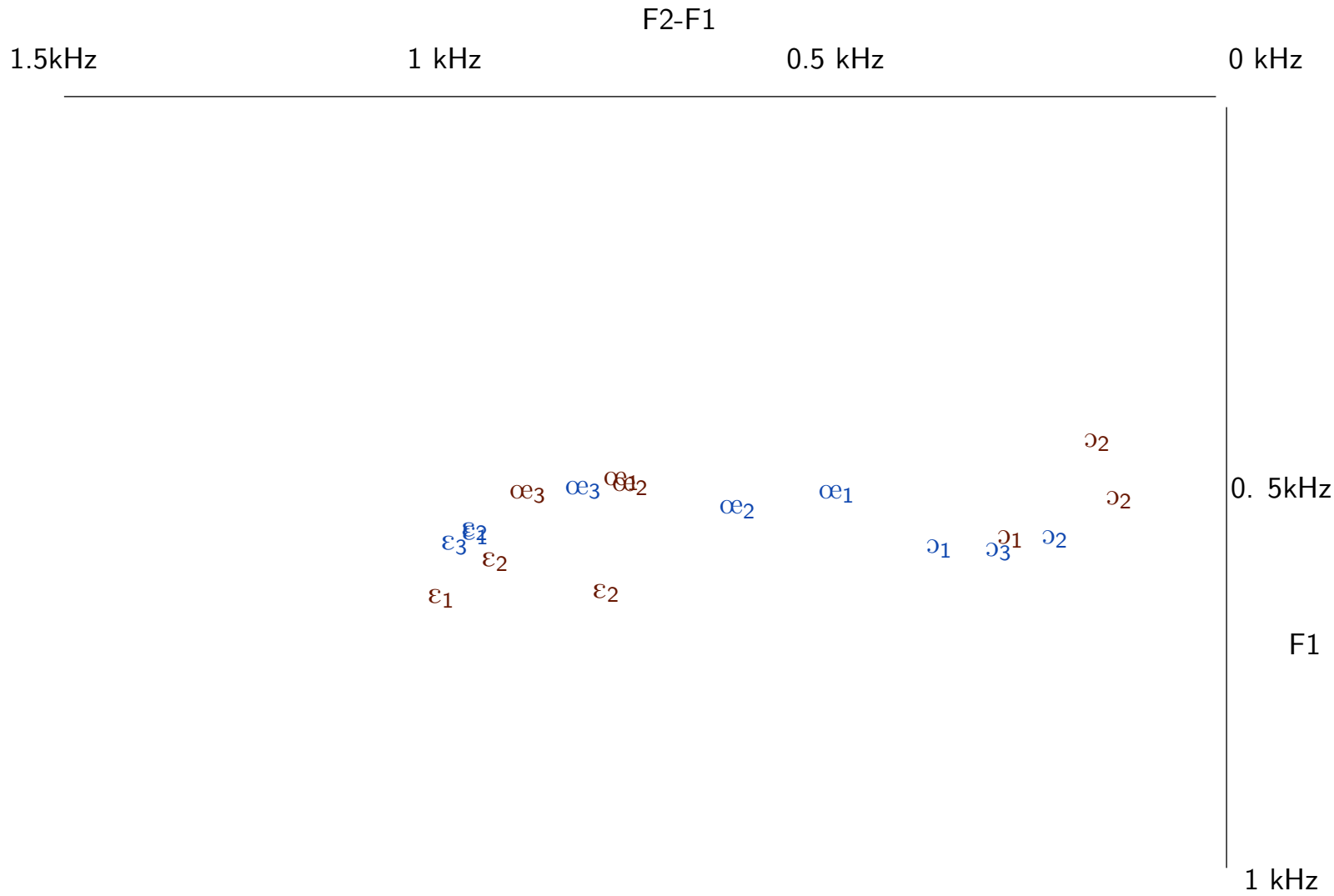


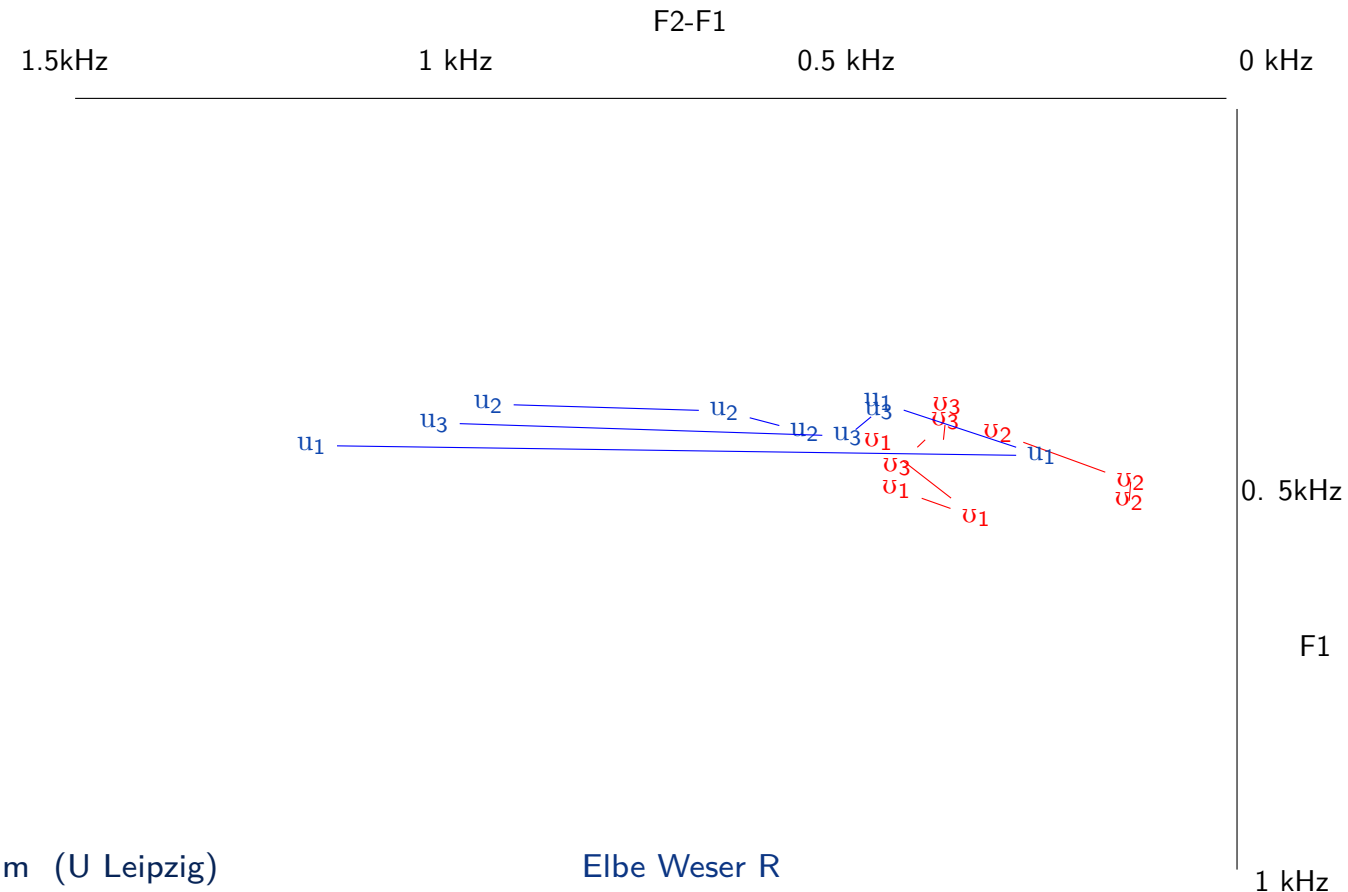
Figure 6: HW mid lax vowels



Diphthongization I

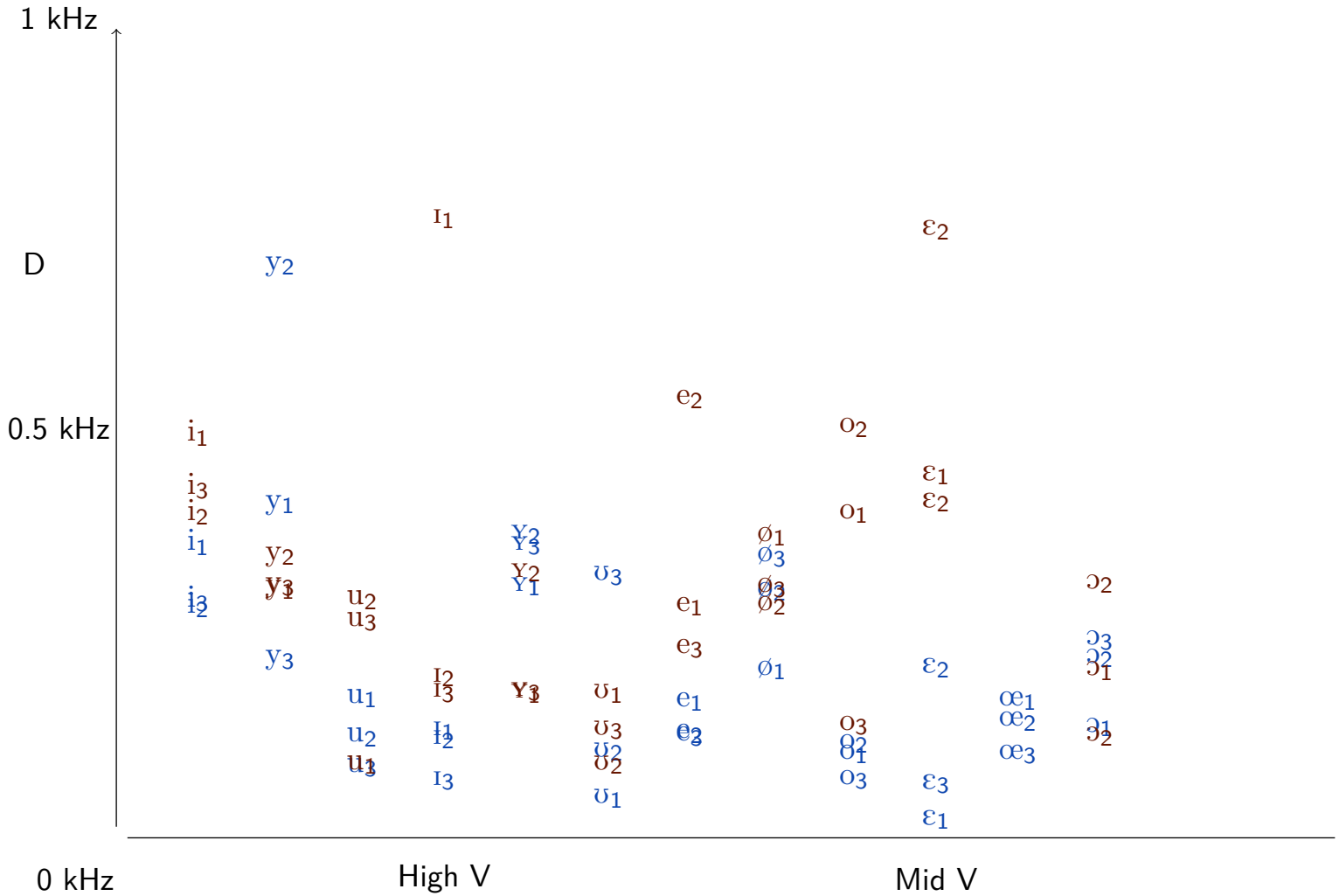
- Total Euclidian Distance (D) between points of measurement calculated.

Figure 7: HW /ʉʁ/ and /ʊʁ/ diphthongization



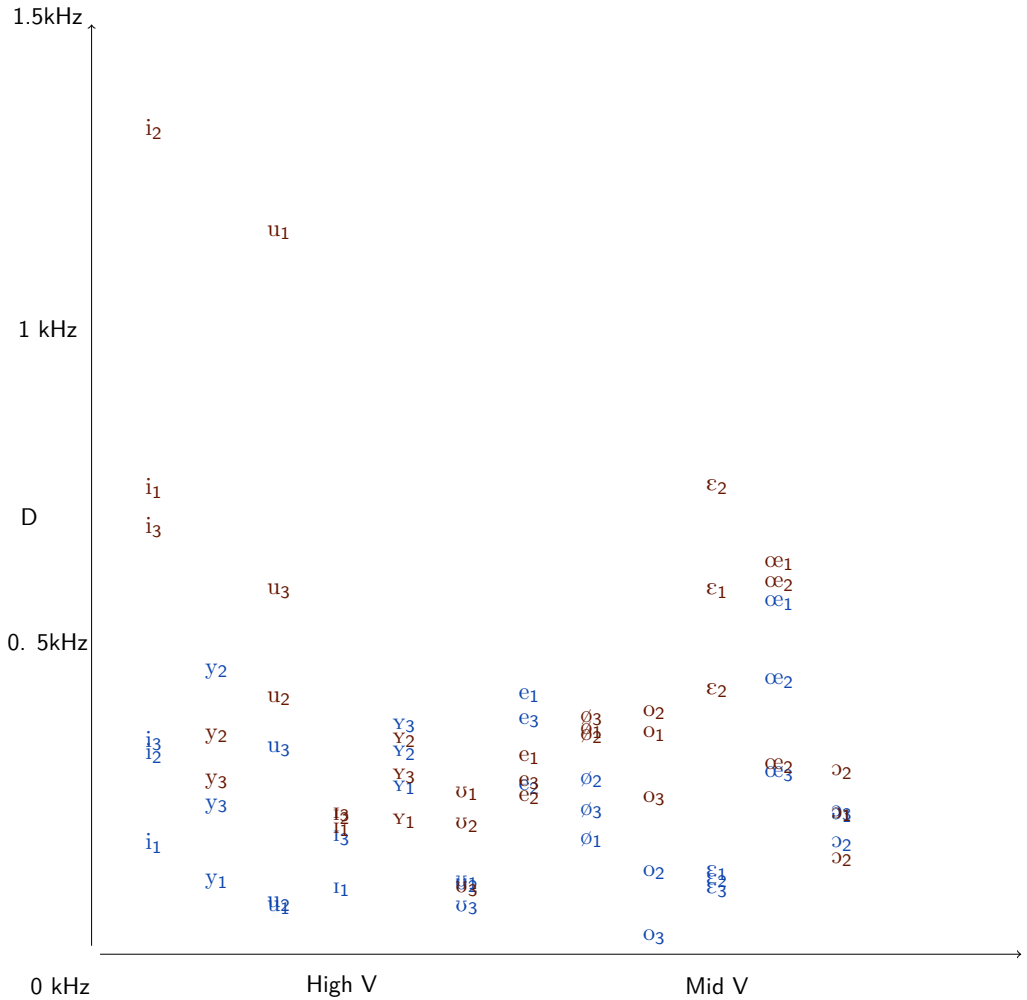
Diphthongization II

Figure 8: Diphthongization Distances TW



Diphthongization III

Figure 9: Diphthongization Distances HW



Summary

The pilot study shows:

- High and mid tense vowels have a higher F2 ($\approx$ are lowered) before coda /ʁ/.
- Tenseness distinctions seems to be neutralized in mid vowels.
- High lax vowels are least likely to surface as diphthongs.
- Tense vowels are more likely to surface as dipthongs.
- This supports our impression that high lax vowels fuse with coda /ʁ/.

Conclusion

- Elbe Weser German coda /ʁ/ fuses with lax high vowels but vocalizes after tense vowels.
- Lax mid vowels optionally dissimilate to tense vowels before vocalized /ʁ/.
- We derive this as an OCP(tense) effect under the assumption that /ʁ/ is underlyingly specified as [–tense].
- This pattern conflicts with the assumption that German tense and lax vowels are only distinguished by length (c.f. i.a. Golston & Wiese, 1998)
- Our pilot study shows that these effects are acoustically real.

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Lax high vowels must fuse

Additional undominated constraint:

- $\textsc{Id}$(t)&(h) Assign $*$ to I-O correspondent segments with different values for [±tense] and [±high].

Lax high vowels must fuse

- *ʏː blocks the unlowered, fused candidate.
- Id(t)&(h) blocks the mid diphthong candidate.

(10) /ʏ1ʁ2/	*ʁ]σ	OCP(-t)	*ʏː	Id(t)&(h)	Id(t)	*œː	*Diph	Uni
a. [ʏ1ʁ2]	*!	*						
☞b. [œː1,2]						*		*
c. [ʏː1,2]			*!					*
d. [øː1e͡ɐ2]				*!		*	*	
e. [ʏ1e͡ɐ2]		*!			*		*	